

Teen Numbers: One Ten and Some Ones

Answer Key

Kindergarten–Grade 1

Quick Answers

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|---------------------------|--|
| 1. the number of ones = 3 | 5. the number = 15 |
| 2. the number of ones = 6 | 6. the number = 19 |
| 3. the number = 14 | 7. (a) the symbol = $>$, — |
| 4. the number = 18 | 8. (a) the correct number of ones = 7, — |
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What is verified

6 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

1.NBT – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–4 of 5 across 10 tagged checks.

- 1.** The number is 13. Fill in: 13 is one ten and _____ ones.

Solution: Take the ten out first. Fill the ten-frame: that uses up ten of the thirteen counters. Count what is left over: thirteen counters take away the ten already in the frame leaves three. So $13 - 10$ is the number of loose ones. 3

- 2.** The number is 16. Fill in: 16 is one ten and _____ ones.

Solution: Same move: the full frame takes ten of the sixteen counters, and $16 - 10$ counters are left outside it. Counting on from ten to sixteen — eleven, twelve, thirteen, fourteen, fifteen, sixteen — takes six steps, which is the same six. 6

- 3.** Kai built one full ten-frame and put 4 loose counters beside it. What number did Kai build?

Solution: Going the other way now: the frame is worth ten, so start at ten and count on the loose counters. Ten, then eleven, twelve, thirteen, fourteen. In notation that is $10 + 4$.

14

- 4.** Mia built one full ten-frame and put 8 loose counters beside it. What number did Mia build?

Solution: Start at ten and count on eight: eleven, twelve, thirteen, fourteen, fifteen, sixteen, seventeen, eighteen. In notation, $10 + 8$. Note the pattern — the ones counter you added is the digit you can hear in the number word. 18

5. Read the model (one full ten-frame plus loose dots) and write the number it shows.

Solution: The full frame is one ten; there are five loose dots beside it. Do not recount the frame dot by dot — that is what the frame is for. Start at ten and count on the five: $10 + 5$.

15

6. Read the model (one full ten-frame plus loose dots) and write the number it shows.

Solution: The full frame is one ten and nine dots sit outside it. Start at ten and count on nine: $10 + 9$. This is the largest teen number — one more counter would fill a second frame and make two tens.

19

7. Both 14 and 12 are built with one full ten-frame and some loose counters. (a) Write $<$, $>$, or $=$ between them. (b) Explain why the loose counters, and not the frames, decide which is greater.

Solution (a): Each number has exactly one full frame, so the frames match. What is left is four loose counters against two loose counters, and four is more.

$14 > 12$

Solution (b) — instructor-judged. A correct response says the tens are equal, so only the ones can decide. Full credit for the idea in any wording: both numbers hold one ten, so the frames cancel out and 4 ones beats 2 ones; or, shown with the models, both have one full frame, and the loose dots settle it; or “they both have a ten so I just looked at the 4 and the 2”. Saying only that 4 is more than 2, without noting that the tens match, earns half credit. Repeating the symbol with no reason earns no credit.

8. Nia says: “17 is one ten and seventeen ones.” (a) Write the correct number of ones. (b) Explain what Nia mixed up.

Solution (a): The ten inside the frame has already been counted, so it must come out of the total before the ones are counted: $17 - 10$.

7

Solution (b) — instructor-judged. A correct response explains the double count. Full credit for the idea that the ten was already counted once inside the frame, so calling all seventeen counters “ones” as well counts those same ten a second time; one ten and seven ones is all seventeen. Accept “she forgot to take the ten out first”, or a drawing of a full frame plus seven loose dots labelled as the whole amount. Saying the answer should be seven ones without saying why earns half credit. Restating the number earns no credit.